Sonya Low

9009 4742 | sonya.low.2025@computing.smu.edu.sg | linkedin.com/in/sonyalow/ | github.com/sonyalow-smu611

Education

Singapore Management University

Bachelor of Science in Information Systems *Aug 2025 – May 2029*

- **Relevant coursework**: Web Application Development, Data Analytics, Data Structures & Algorithms, Database Management Systems, Statistics

Experience

Business Intelligence and Analytics Club's AI Lodge Programme Jan 2026 – June 2027

Head of AI Lodge Programme

- Led and managed programme covering ML fundamentals, RAG, AI agents, red teaming and deployment, equipping mentees to build end-to-end AI applications.
- Built and deployed interactive marketing website for AI Lodge programme
- Designed and built a working reference project for every curriculum module (RAG, multi-agent systems, LLM red teaming) for mentees to build alongside, supporting 70+ completed AI projects.
- Recruited 70 participants from 200+ applicants, collaborating with 20+ mentors, and organised the programme's AI Hackathon to showcase technical projects and solutions.

Lodge Mentor

- Taught Intro to ML and AI theory, covering ML Learning methods, algorithms, Neural Networks, Attention mechanism and Transformer model.
- Led workshop on building a multi-agent workflow with Langchain and deployment on Langgraph.
- Led workshop on building a LLM-as-a-judge agent for LLM Evals, facilitated red-teaming exercise.
- Led workshop on implementing RAG in an agentic pipeline.

Lodge Mentee

- Vibecoded final Lodge Programme individual project – AI News and Current Affairs video generator pipeline which is an agentic pipeline automatically creating shortform vertical videos on top headlines from past 2 weeks.

Fintech Innovators Hackathon March 2026

Placed 3rd out of 200 teams

- Built a macroeconomic news intelligence platform, implementing a HeatScore ranking algorithm and associations graph to surface and connect high-impact economic events in real time
- Built an interactive associations graph (D3.js force-directed, React) that scores every article with a weighted algorithm, enabling analysts to trace connected events without shared keywords.
- Built a RAG-powered AI chat assistant (OpenAI, Supabase pgvector) that embeds each query and retrieves the most relevant articles and persisting per-user chat history to give analysts continuous institutional memory across sessions.

Jane Street WiSE Programme Hong Kong July 2025

In cohort of 70+ women from STEM

- Learned trading and risk management from Jane Street traders and strategy and product team
- Learned how math and computational thinking is used to problem solve and make decisions in a Quantitative Trading firm.

Projects

Hoppers - Cafe Discovery App | *Next.js, Express, React, PostgreSQL, REST API* May 2026 - July 2026

- Built a full-stack café discovery app with a decoupled REST API, enabling independent frontend/backend deployment.
- Built a real-time data aggregation system processing live user-submitted metrics into summarized status indicators, replacing static listing data.
- Integrated an LLM-powered recommendation pipeline (OpenAI SDK) with structured prompt design, returning ranked, scored results from real-time user input.

Roommate Finder | *MERN Stack* Feb 2026 - April 2026

- Built a complete MVC web app structured as a full Express + MongoDB (Mongoose) application split cleanly into models, controllers, views and router, with middleware authentication and authorisation.
- Learned to design RESTful, resource-oriented routing, splitting routes into dedicated route files, each mapping HTTP path to a specific controller function and view
- Learned to wire cross-cutting concerns through the MVC layers instead of duplicating them, setting session state once at login and reuse it as an auth guard across every controller, and exposed globally to all views

LearnActively – OpenAI Build Week 2026 | *React, Next.js, Zod, Drizzle, Supabase* June 2026 - July 2026

- Built an AI-powered active-learning workspace. It turns any topic into a structured learning session with a visual overview, cheatsheet, concept roadmap, node-by-node explanations, practice labs, flashcards, and a quiz.
- Designed a multi-agent orchestration pipeline with 8+ specialized agents (planner, concept-tree, quiz, flashcard, etc.) producing schema-validated structured outputs (Zod) rendered as interactive UI, including ReactFlow roadmaps and Mermaid diagrams.
- Implemented real-time streaming and persistence architecture using Server-Sent Events, Drizzle ORM with Supabase Postgres, and tiered model routing (fast/reasoning/structured) with resilient fallback generation for offline demo reliability.

Langchain Vibecoding Agent Workflow | *Python, Langchain, Pydantic* July 2026

- Designed a workflow where a product request flows through distinct agent roles: a director, an architect, and frontend/backend workers that generate implementation code, orchestrated as a graph rather than a single linear chain.
- Used Pydantic schemas to force LLM outputs into typed, structured formats (specs, tasks, architecture, test reports, repair tickets) and defined shared graph state so each agent's output cleanly feeds the next stage

Customer Segmentation and Marketing Analytics — Reverie Hacks Datathon Track 2026 Aug 2026

- Built an end-to-end unsupervised learning ML pipeline to segment 2,237 customers into 4 behavioral personas using K-Means clustering on 10 engineered features
- Engineered a leakage-safe train/validation/test split pipeline, with optimal cluster count (k=4), validated cluster stability across train/val/test splits to prevent overfitting.

Tiktok Hackathon Aug 2026

- Co-developed a hybrid AI retrieval pipeline combining BM25, Sentence-Transformer embeddings, and constraint-based ranking, improving retrieval performance of RAG system.
- Engineered and evaluated a multi-turn conversational AI agent with session-state memory, intent routing, slot extraction and adaptive clarification.

Technical Skills

Languages: Python, SQL(Postgres), JavaScript, HTML, CSS
Frameworks: React, Bootstrap, Vue.js, Node.js, Flask, FastAPI, Next.js
Developer Tools: Git, Docker, Google Cloud Platform, PyCharm, IntelliJ, Eclipse
Agentic Tools: n8n, Langchain/LangGraph/Langsmith, Zapier, Make.com, n8n
Libraries: pandas, NumPy, Matplotlib